Tutorial - 9

① $D(\alpha) = e^{\alpha a^\dagger - \alpha^* a}$

(1)

$$D^\dagger(\alpha) = \cancel{e^{\alpha a^\dagger - \alpha^* a}} \; e^{\alpha^* a - \alpha a^\dagger}$$

$$\therefore D^\dagger D = \cancel{e^{\alpha a^\dagger - \alpha^* a}} \; e^{\alpha a^\dagger - \alpha^* a} \cdot e^{\alpha a^\dagger - \alpha^* a}$$

From reduced BCH:

$$e^A \cdot e^B = e^{A + B + \frac{1}{2}[A, B]}$$

$$\therefore A = \cancel{\alpha a^\dagger - \alpha^* a} \; \alpha^* a - \alpha a^\dagger \; ; \; B = \alpha a^\dagger - \alpha^* a = -A \longrightarrow \boxed{[A, B] = 0}$$

Expt. No. _____ Page No. _____

$\therefore \; D^\dagger D = e^0 = \mathbb{I}$

$\therefore \; \langle m, \alpha | n, \alpha \rangle = \cancel{D\,m} \; \langle m | D^\dagger D | n \rangle = \langle m | n \rangle = \delta_{mn}$

$\therefore \; \boxed{\langle m, \alpha | n, \alpha \rangle = \delta_{mn}}$ (Proved)

(ii) $|n, \alpha\rangle = D|n\rangle. \; \cancel{= D}$ — (i)

$|0, \alpha\rangle = D|0\rangle.$

$|n\rangle = \dfrac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle.$ — (ii)

$\therefore \; |n, \alpha\rangle = D \dfrac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle.$

$= \cancel{D(\;D}$

$D(a^\dagger)^n = D a^\dagger a^\dagger \ldots\ldots$ n times.

$= D a^\dagger (D^\dagger D) a^\dagger (D^\dagger D) \ldots\ldots$ $[D^\dagger D = \mathbb{I}]$

$= (D a^\dagger D^\dagger)(D a^\dagger D^\dagger) \ldots\ldots D$ n times.

$\boxed{D(a^\dagger)^n = (D a^\dagger D^\dagger)^n D}$. — (iii)

$\therefore \; |n, \alpha\rangle = \dfrac{(D a^\dagger D^\dagger)^n}{\sqrt{n!}} D |0\rangle. = \dfrac{(D a^\dagger D^\dagger)^n}{\sqrt{n!}} |0, \alpha\rangle.$

$D^\dagger a^\dagger D = a^\dagger + \alpha^*.$

$\Rightarrow \; D(D^\dagger a^\dagger D) D^\dagger = D(a^\dagger + \alpha^*) D^\dagger$

$\Rightarrow \; a^\dagger = D a^\dagger D^\dagger + \alpha^*$ $[D D^\dagger = \mathbb{I}]$

$\Rightarrow \; D a^\dagger D^\dagger = a^\dagger - \alpha^*$ — ~~(iv)~~ (iv)

Using this : $\boxed{|n, \alpha\rangle = \dfrac{(a^\dagger - \alpha^*)^n}{\sqrt{n!}} |0, \alpha\rangle}$ (Proved) — (v)

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(iv) (a) $[A, A^\dagger] = -[a^\dagger - \alpha^*, a - \alpha]$.

$= -[a^\dagger, a] = 1$ (Proved)

(b) $A^\dagger |n, \alpha\rangle = (a^\dagger - \alpha^*) |n, \alpha\rangle$

$= (a^\dagger - \alpha^*) D |n\rangle$

$D^\dagger a^\dagger D = a^\dagger + \alpha^*$

$\Rightarrow$ ~~Dta~~ $a^\dagger D = D a^\dagger + \alpha^* D$.

$\Rightarrow a^\dagger D - D a^\dagger = \alpha^* D \Rightarrow \boxed{[a^\dagger, D] = \alpha^* D}$ — (vi)

$\Rightarrow a^\dagger D = \alpha^* D + D a^\dagger$.

Using this :

$A^\dagger |n, \alpha\rangle = (a^\dagger D - \alpha^* D) |n\rangle = \alpha^* D a^\dagger |n\rangle$.

$= D \sqrt{n+1} |n+1\rangle$.

$\boxed{A^\dagger |n, \alpha\rangle = \sqrt{n+1} |n+1, \alpha\rangle}$. (Proved) — (vii)

$\therefore A |n, \alpha\rangle = (a - \alpha) D |n\rangle$

$= (aD - \alpha D) |n\rangle$

$D^\dagger a D = a + \alpha \Rightarrow \boxed{aD = Da + \alpha D}$ — (viii)

$\therefore A |n, \alpha\rangle = Da |n\rangle = \boxed{\sqrt{n} |n-1, \alpha\rangle = A |n, \alpha\rangle}$ (Proved) — (ix)

(2) (i) $x = r \sin\theta \cos\phi$; $y = r \sin\theta \sin\phi$; $z = r \cos\theta$.

$(x, y, z) \longrightarrow (-x, -y, -z)$.

$\therefore \sin\theta \cos\phi \longrightarrow -\sin\theta \cos\phi$; $\cos\theta \longrightarrow -\cos\theta$

$\sin\theta \sin\phi \longrightarrow -\sin\theta \sin\phi$

Date

Expt. No. Page No.

These are equivalent to the following angle transformat^n:

$$\theta \to \pi - \theta \quad ; \quad \phi \to \phi + \pi$$

$r = \sqrt{x^2 + y^2 + z^2}$ remains unchanged.

$$\therefore \boxed{\psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi)} \quad \text{(Ans)}$$

(ii) $x = \rho \cos\phi \quad ; \quad y = \rho \sin\phi \quad ; \quad z = z$

$(x, y, z) = (-x, -y, -z)$.

$\therefore \cos\phi \to -\cos\phi \quad ; \quad \sin\phi \to -\sin\phi \quad ; \quad z \to -z$

This is equivalent to the ~~angle~~ transformat^n:

$$\phi \to \phi + \pi \quad ; \quad z \to -z$$

$\rho = \sqrt{x^2 + y^2}$ remains unchanged.

$$\therefore \boxed{\psi(\rho, \phi, z) = \psi(\rho, \phi + \pi, -z)} \quad \text{(Ans)}$$

(iii) (a) $P\psi_1 = -A(x + y + z) \exp(-(x^2 + y^2 + z^2))$

$= -\psi_1 \to$ odd parity.

(b) $P\psi_2(r, \theta) = \psi_2(r, \pi - \theta)$

$= -Br \exp(-r^2) \cos\theta = -\psi_2(r, \theta)$

[odd parity]

(c) $P\psi_3(\rho, \phi, z) = \psi_3(\rho, \phi + \pi, -z) = +c \dfrac{\sqrt{\rho^2 + z^2}}{z^5} \sin\phi \cancel{= \psi_3}$

$= \psi_3(\rho, \phi, z)$

[even parity]

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